

# SUSHANT LOKHANDE

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## EXPERIENCE

**San Jose State University** | Graduate Research Assistant, Retrieval & ML Aug 2025 to May 2026

- Built and evaluated **RAG** pipelines (chunking, embeddings, vector retrieval, top-k, grounded prompts), separating **retrieval** failures from generation failures to guide fixes.
- Ran **evals** on RAG workflows by comparing chunking strategies, embedding models, and top-k settings, and classified each failure as retrieval-caused or generation-caused.
- Cut experiment setup effort by about **55%** by automating orchestration, configuration, **telemetry** collection, and analysis, making large experiment sweeps reproducible.
- Supported **200+ students** as a TA in Programming Paradigms, AI, and Machine Learning through debugging sessions, code review, and grading.

**Turito** | Software Engineering Intern, Product Reliability Jun 2022 to Oct 2022

- Cut controlled-disruption failures by **42%** (3.1% to 1.8%) by hardening **retries**, timeouts, failure recovery, and distributed state transitions.
- Slashed p95 **CPU** by **22%** by profiling high-frequency media paths and removing excess buffer allocation, payload copies, repeated routing lookups, and **lock contention**.
- Reduced production diagnosis time by **35%** by adding structured **metrics**, runtime-state visibility, and targeted logs.
- Worked across a **WebRTC** control plane and **RTP/RTCP** media paths spanning about **12 regions**, including ICE/STUN/TURN connection setup and degraded-network handling.

**Trivia Softwares** | Software Engineering Intern Nov 2021 to Jan 2022

- Built a **Rust/Tokio** load-testing framework driving about **25,000** concurrent sessions, exposing lock contention, serialized **WebSocket** writes, and slow session cleanup.
- Reduced performance-investigation time by about **40%** with tooling built on Rust workers, **Python** aggregation, **SQLite**, Flask, and Chart.js.

## TECHNICAL SKILLS

**Languages:** Python, TypeScript, JavaScript, C++, SQL

**LLM & AI:** RAG, embeddings, evals, agents, tool use, LangChain, OpenAI and Anthropic APIs, PyTorch, Hugging Face

**Backend & Frontend:** FastAPI, asyncio, SSE, REST, gRPC, React, Next.js, Node.js

**Data & Infra:** PostgreSQL, Redis, Qdrant, SQLite, Kafka, Temporal, Docker, AWS, pytest

## PROJECTS

**AI Learning Hub, Grounded RAG** | *Python, FastAPI, LangChain, OpenAI, Docker, AWS*

- Built grounded **RAG** over **10,000+** articles with source citations and abstention when retrieved evidence was insufficient.

**Relay, LLM Inference Gateway** | *Python, FastAPI, asyncio, SQLite, SSE*

- Built an OpenAI-compatible gateway with exact and **semantic caching**, provider **failover**, and rate limiting that cut p50 latency from **759 ms to 44 ms** over a 4,000-request load test.

**ReelRank, Two-Stage Recommender** | *PyTorch, C++ HNSW, FastAPI, React, TypeScript*

- Built a two-tower retrieval model with ANN candidate generation and a neural ranker, served through **FastAPI** and a **React/TypeScript** UI with metadata-aware cold start.

## OPEN SOURCE

- **microsoft/markitdown**: fixed EPUB conversion for percent-encoded manifest paths, with regression tests (merged)
- **NVIDIA/garak**: preserved attempt UUIDs in report aggregation for traceable LLM evals (merged, shipped in v0.17.0)

## EDUCATION

**San Jose State University** | M.S. in Computer Science | GPA 3.8/4.0 | Spartan Scholar Aug 2024 to May 2026

*Coursework:* Machine Learning, Artificial Intelligence, Distributed Computing, Database Systems, Advanced Algorithms, NoSQL

**University of Mumbai** | B.Tech. in Computer Engineering | GPA 3.9/4.0 Jun 2019 to May 2023

*Research:* malware image-classification ensemble selected for Dr. Mark Stamp's ML textbook; sole-author contactless palmprint verification (0.9975 AUC)